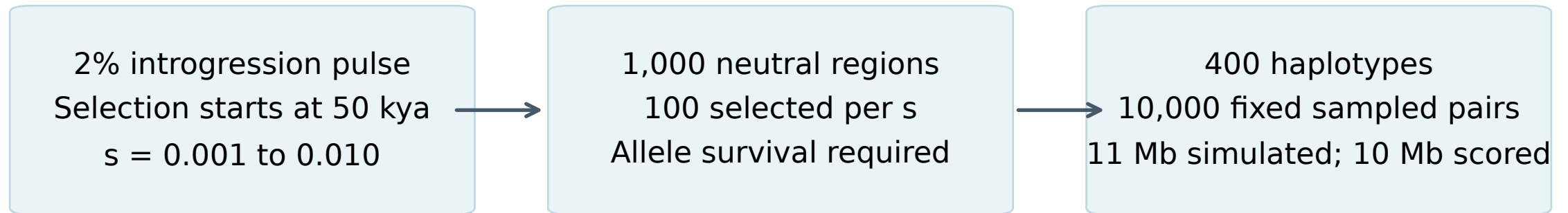


An archaic-carrier coalescence scan

Completed simulation design and the primary score



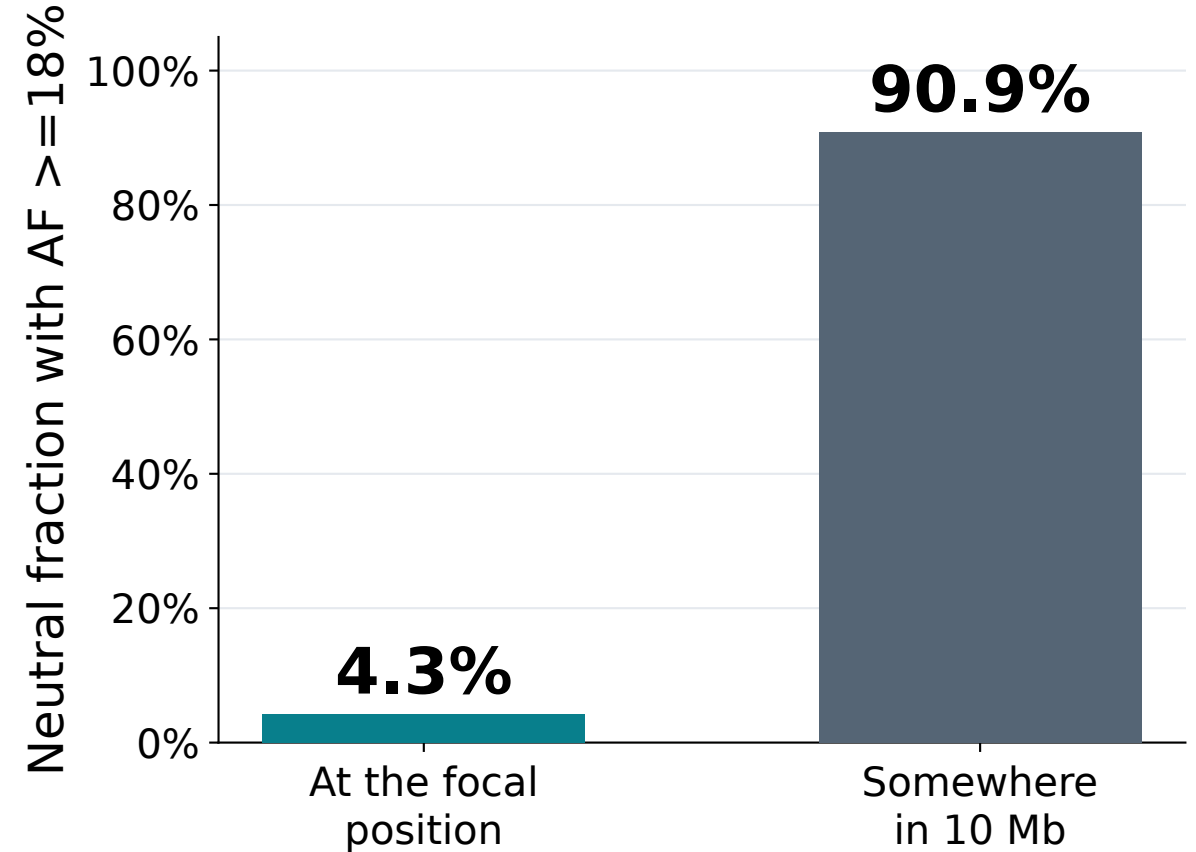
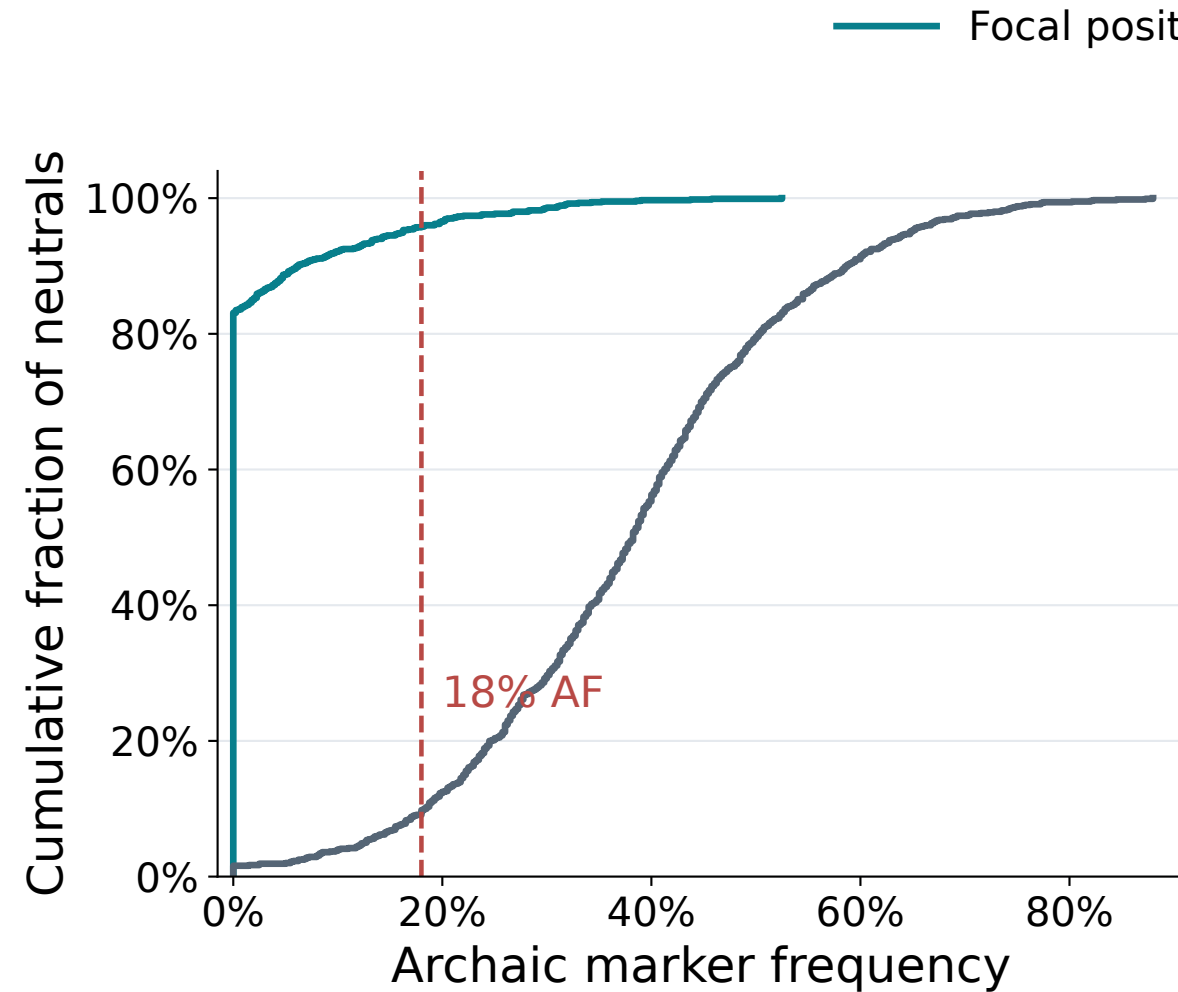
$$S_T = \text{frac_recent}_T(\text{ALT/ALT}) \times \frac{n_{\text{ALT/ALT}}}{n_{\text{all pairs}}} = \frac{n_{\text{recent ALT/ALT}, T}}{n_{\text{all pairs}}}$$

Eligible archaic ALT: oracle-specific origin after the archaic-modern split, before the pulse.

Gamma-SMC frac_recent_T uses posterior-mean TMRCA hard calls.

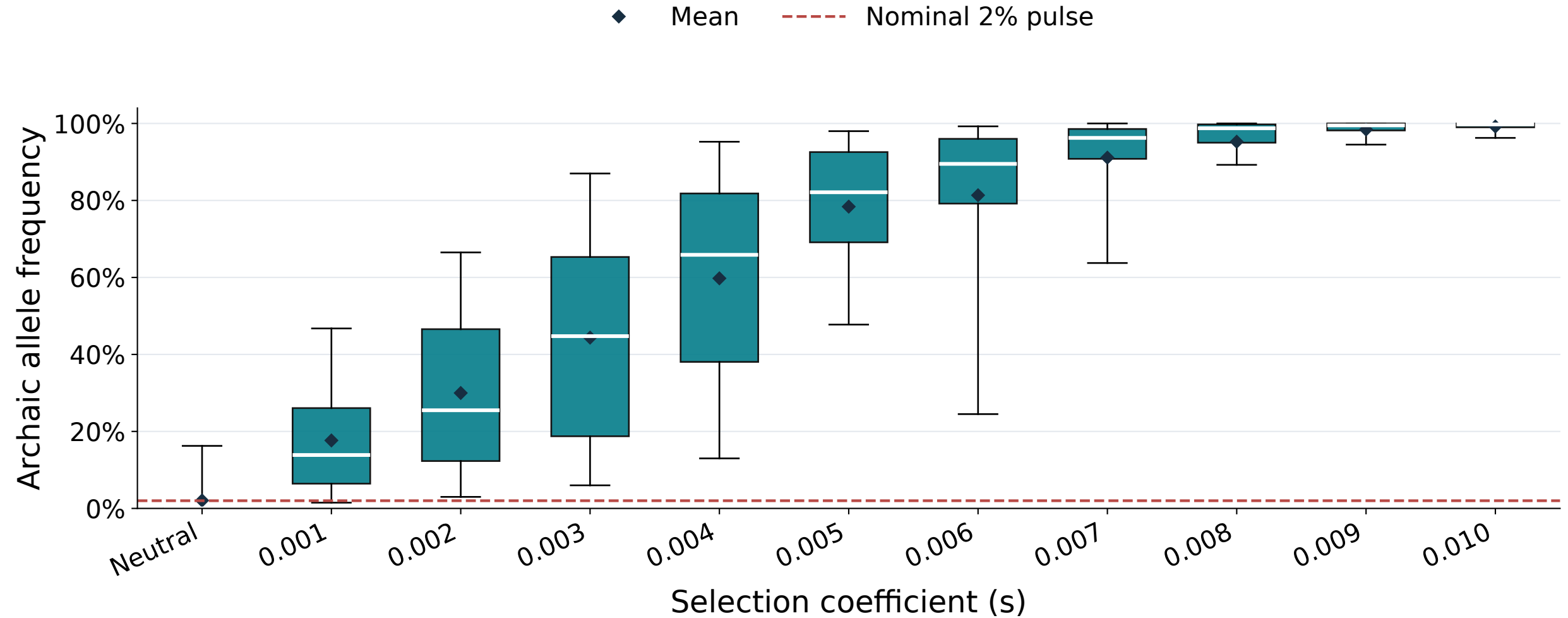
The null must match the position being tested

Same 1,000 neutral simulations; different spatial questions



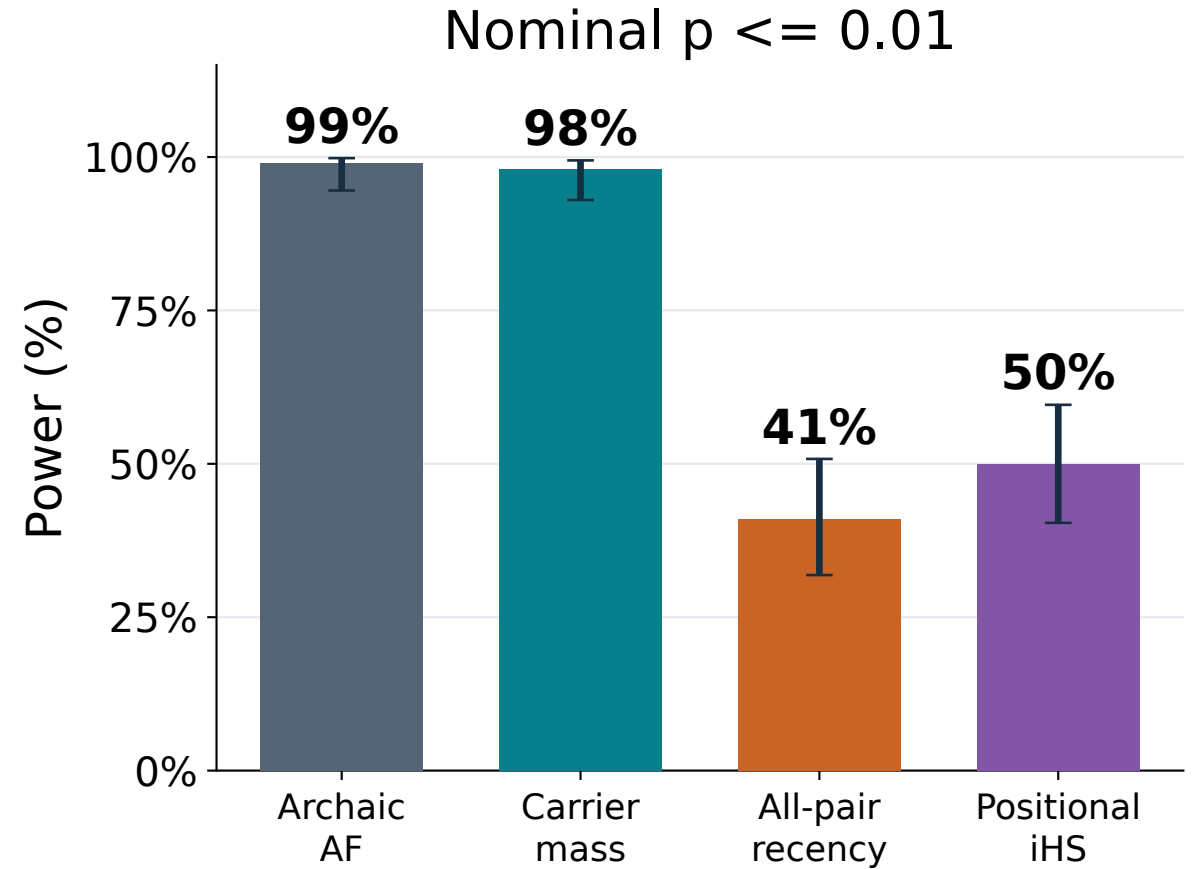
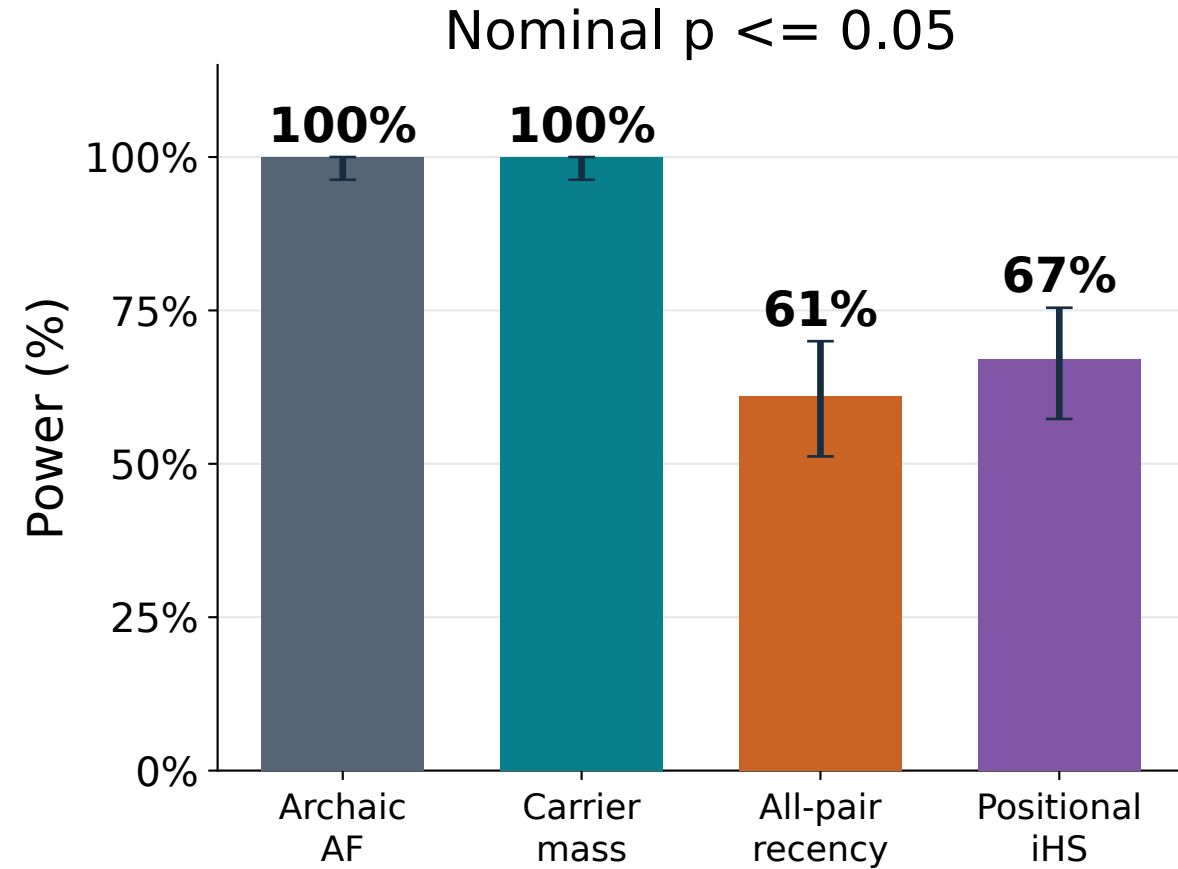
Selection shifts local archaic-allele frequency

Completed cohort: 100 selected replicates per coefficient; 1,000 neutral focal positions



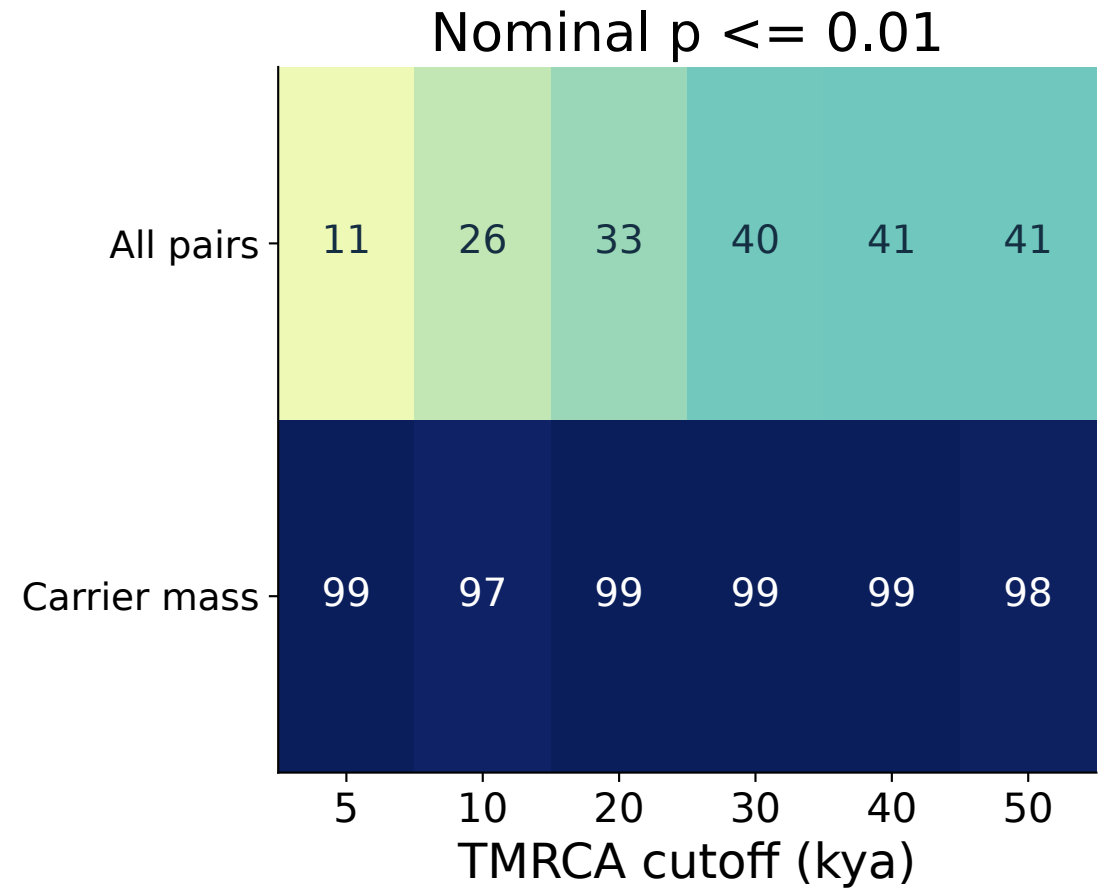
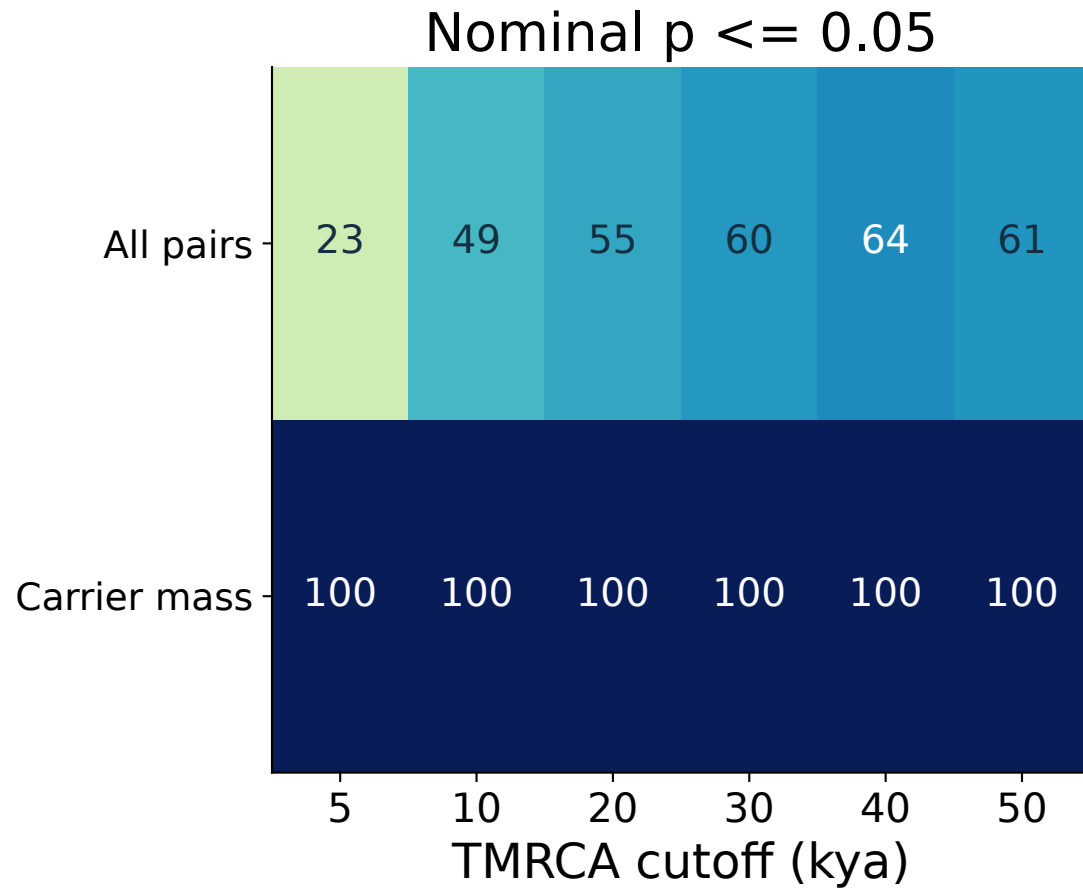
Gamma-SMC power at $s = 0.005$

100 selected + 1,000 neutral regions | $T = 50$ kya | error bars: 95% Wilson intervals



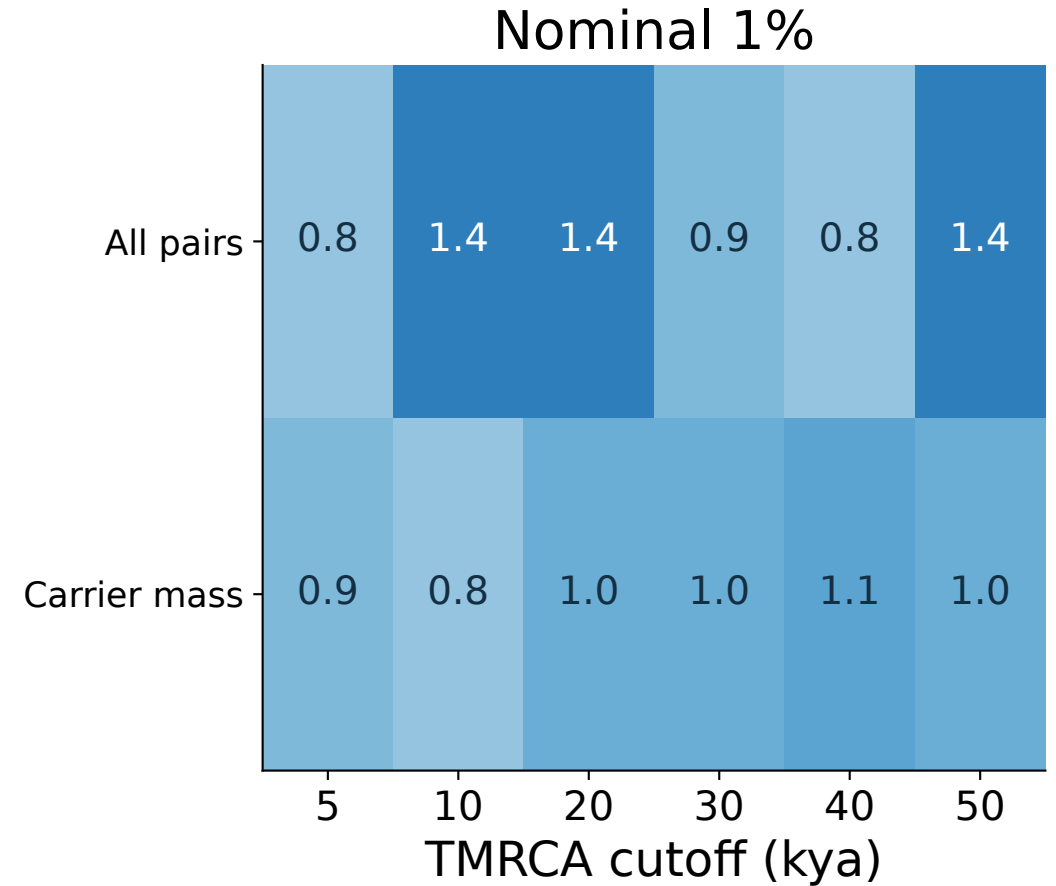
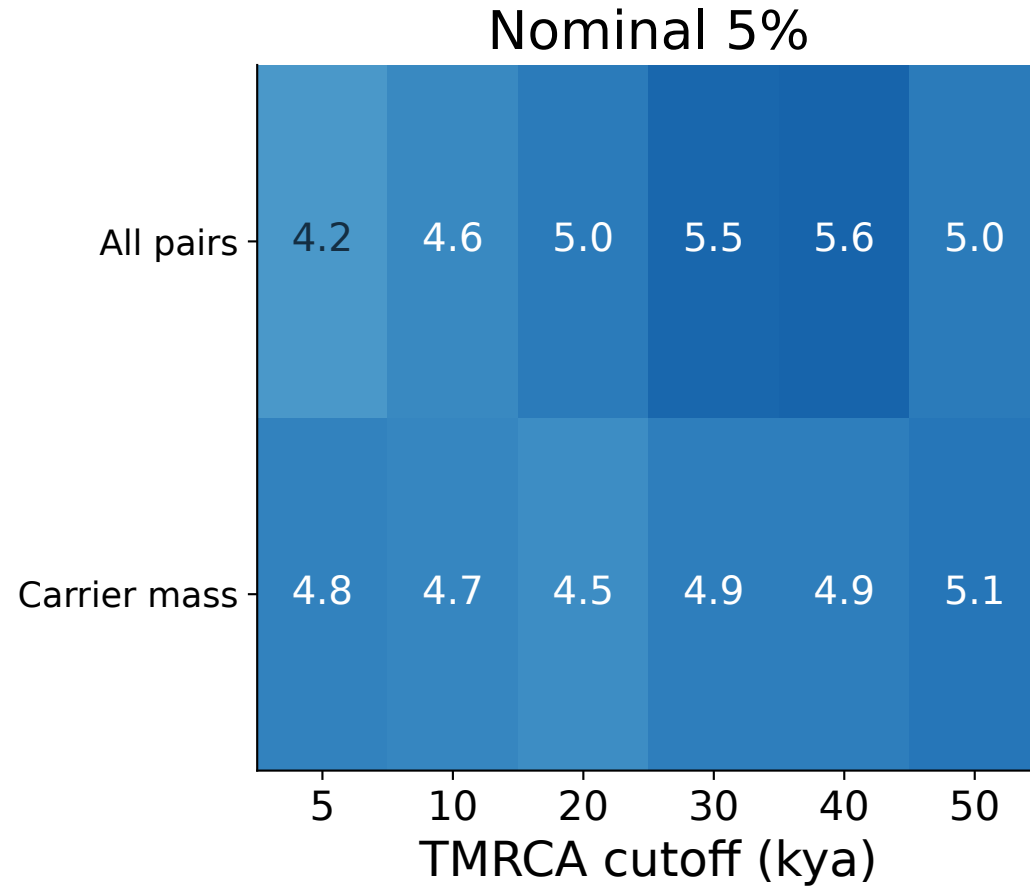
TMRCA cutoffs change decoded power

Gamma-SMC | $s = 0.005$ only | 100 selected replicates per cell | values are percentages



Gamma-SMC positional false-positive rates

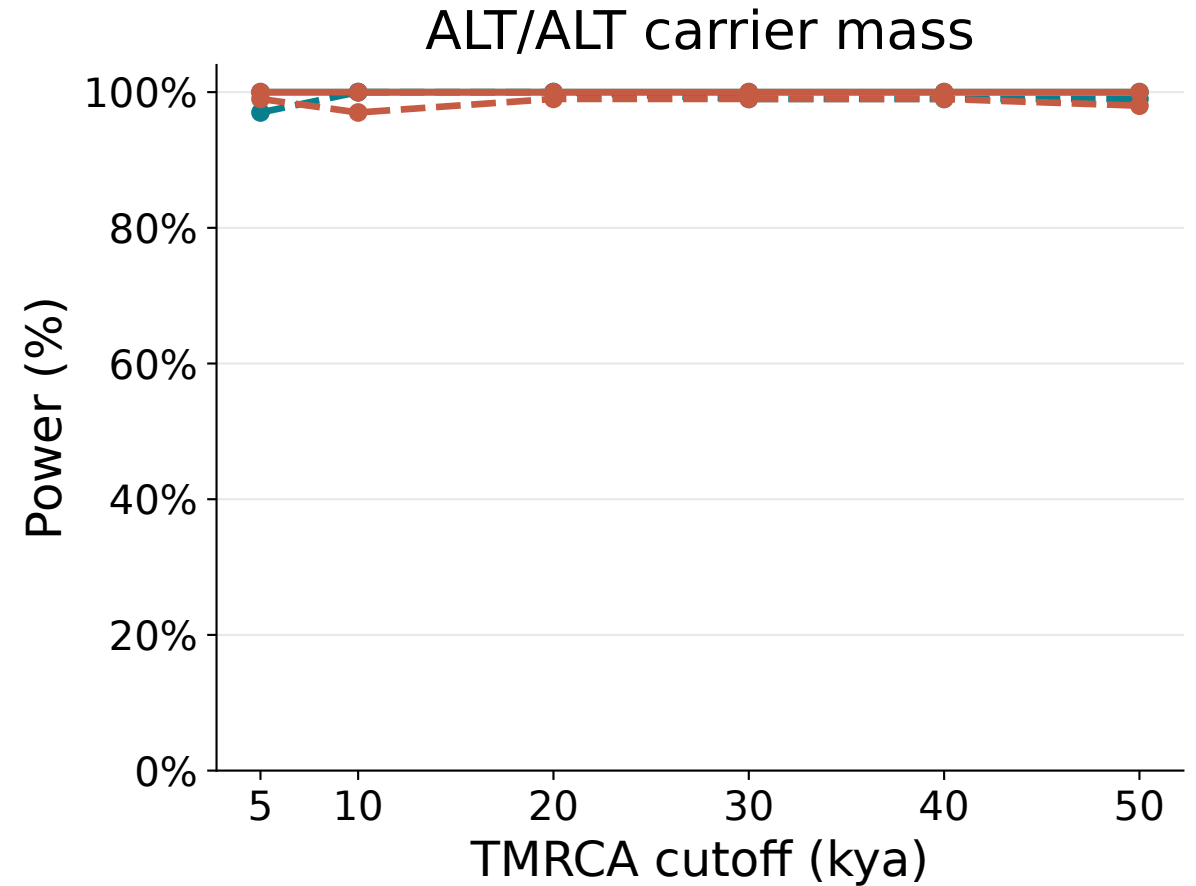
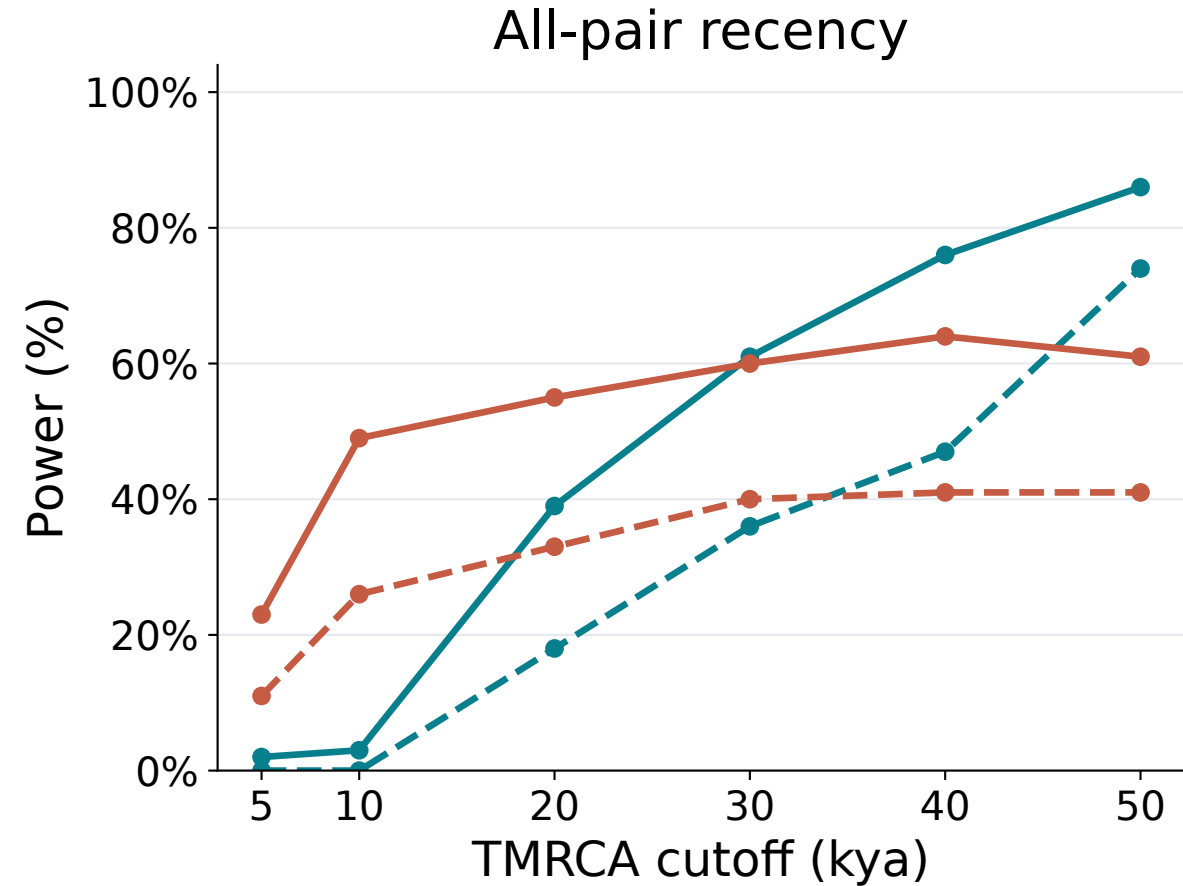
Decoded statistics | 1,000 neutral positions tested once each | FPR, not discovery FDR



Validation: truth versus decoded TMRCA

$s = 0.005$ only | explicit truth reference comparison | 100 selected + 1,000 neutral regions

—●— Truth, $p \leq 0.05$ -●- Truth, $p \leq 0.01$ —●— Decoded, $p \leq 0.05$ -●- Decoded, $p \leq 0.01$

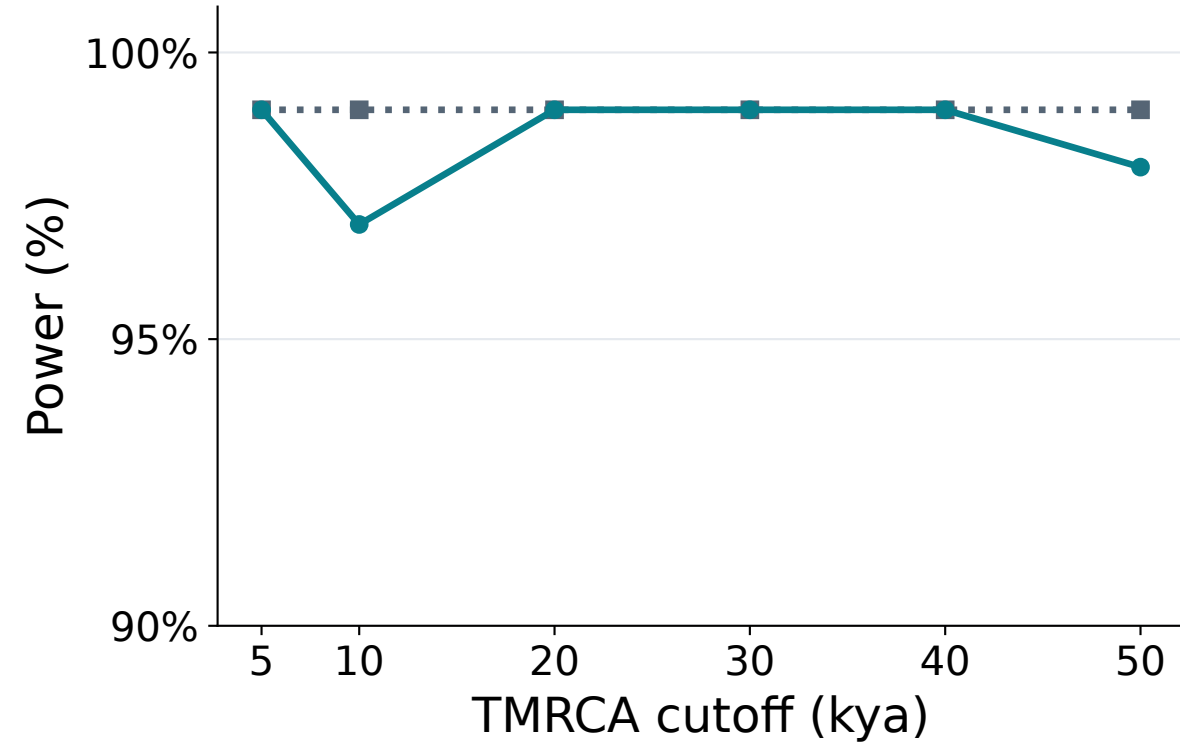


Decoded carrier mass has little gain over AF here

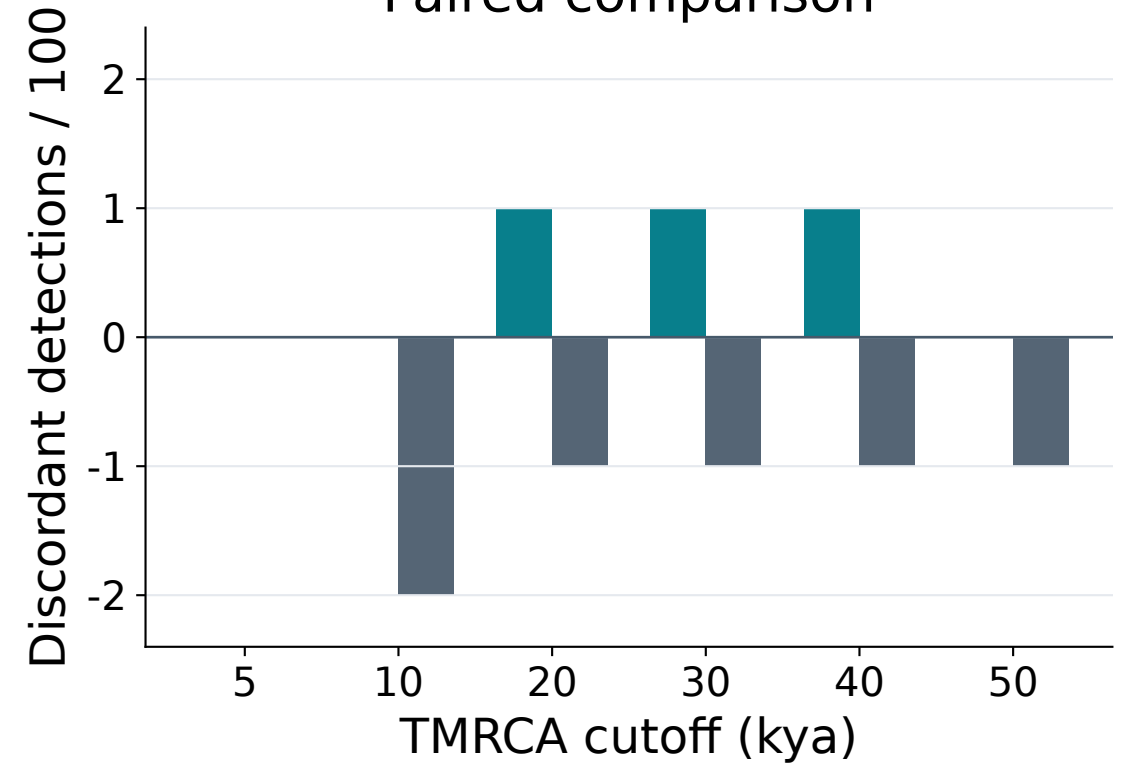
s = 0.005 | nominal p ≤ 0.01 | weaker-selection arms are not decoded

··■·· Archaic AF ■ Carrier only (+)
—●— Decoded carrier mass ■ AF only (-)

Power (zoomed scale)



Paired comparison

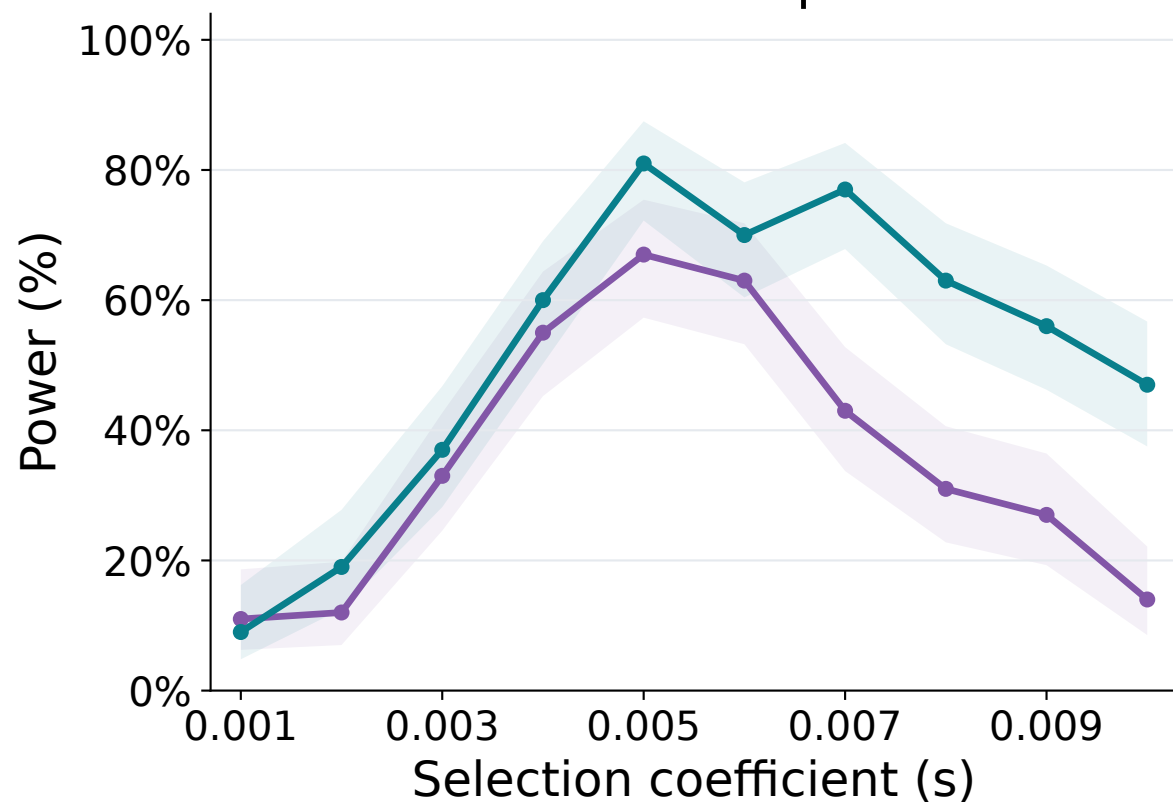


iHS is strongest before the target approaches fixation

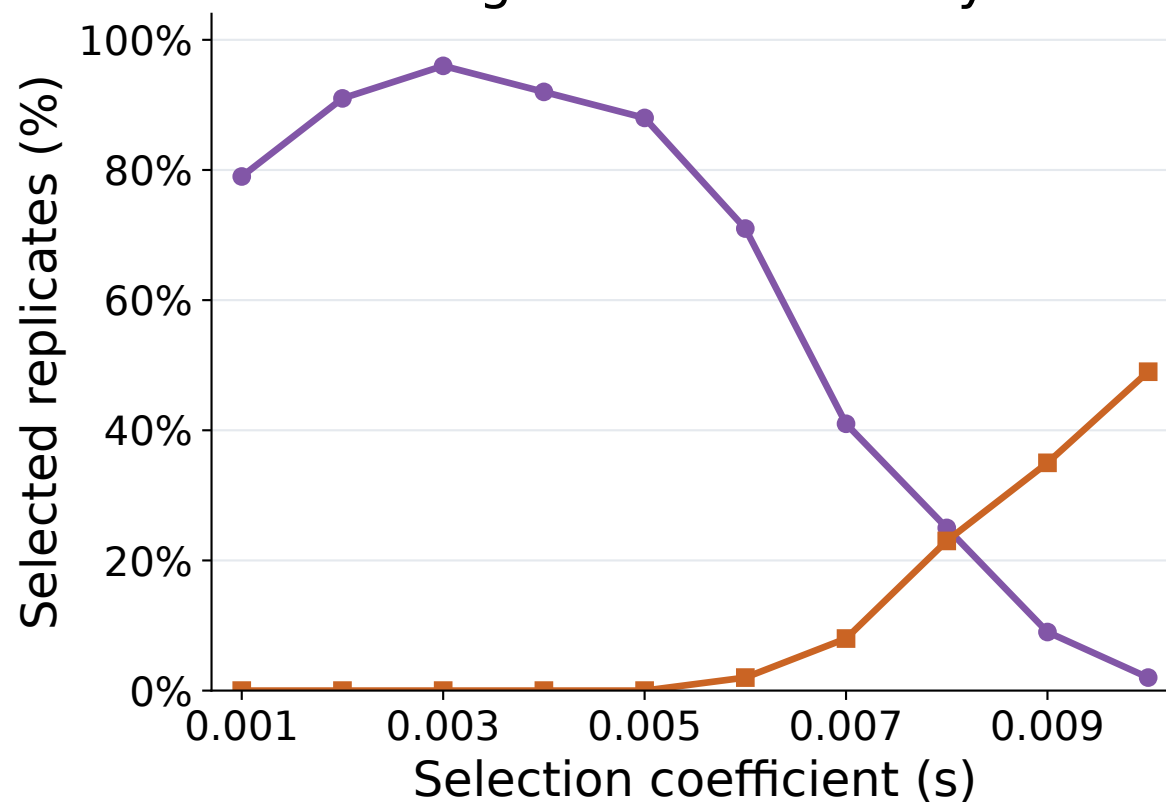
Conventional frequency-standardized iHS | all selected replicates retained

- Positional iHS
- Fixed 100-kb window
- Focal SNP has finite iHS
- Selected allele fixed

Power at nominal $p \leq 0.05$



Target-site availability

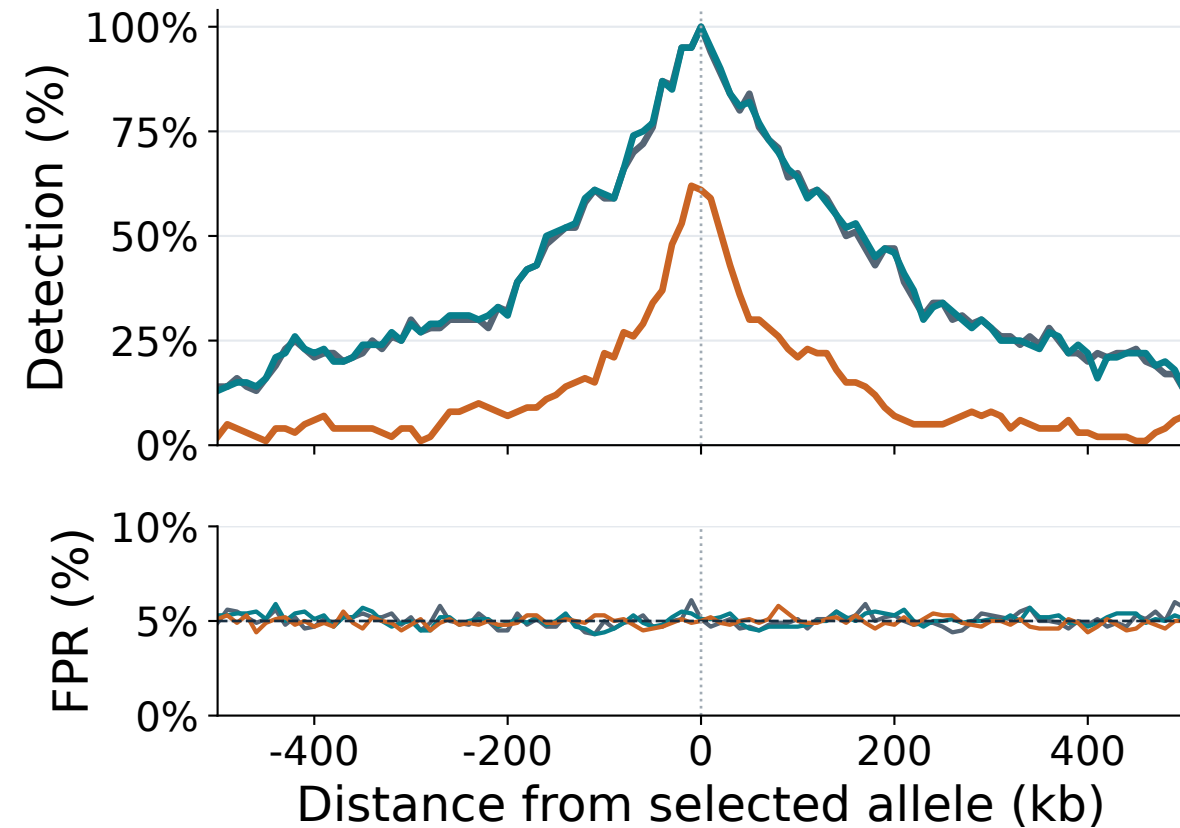


The decoded signal decays with distance

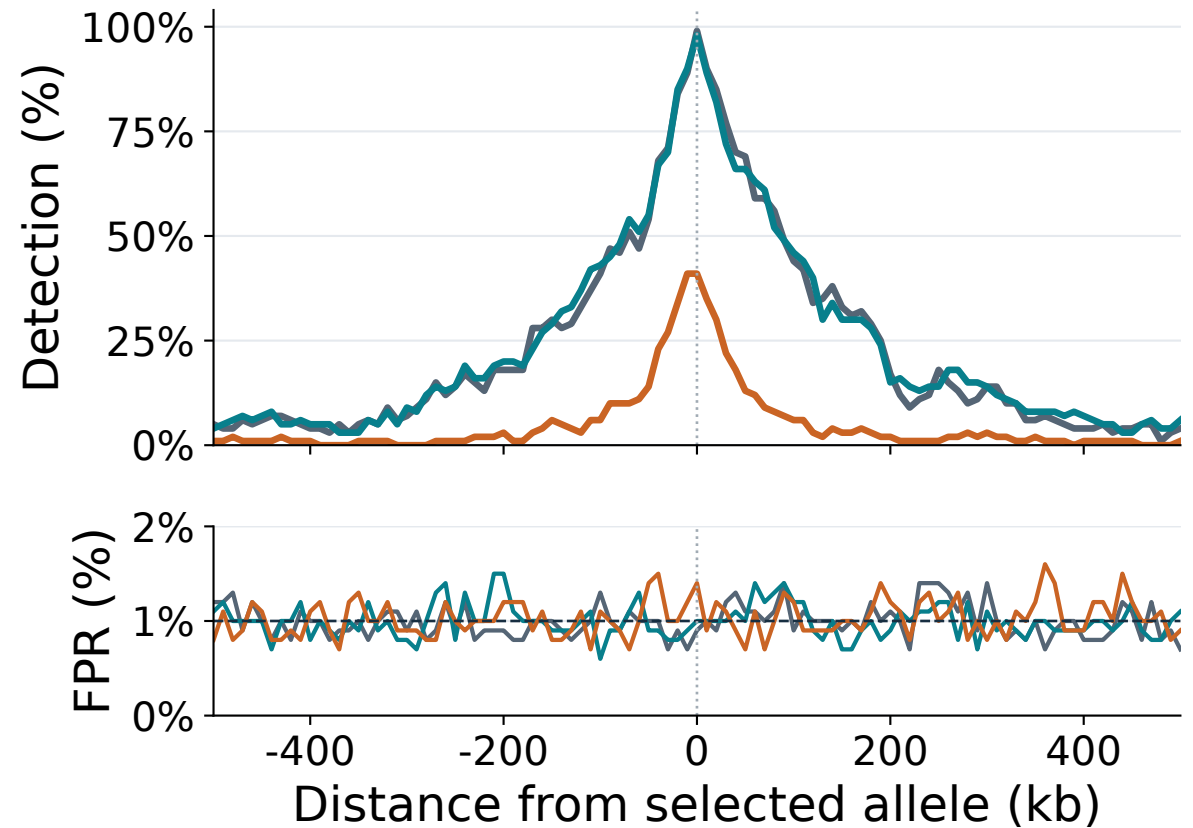
Gamma-SMC | $s = 0.005$; $T = 50$ kya | every stride uses its own matched decoded null

— Archaic AF — Carrier mass — All-pair recency

Nominal $p \leq 0.05$



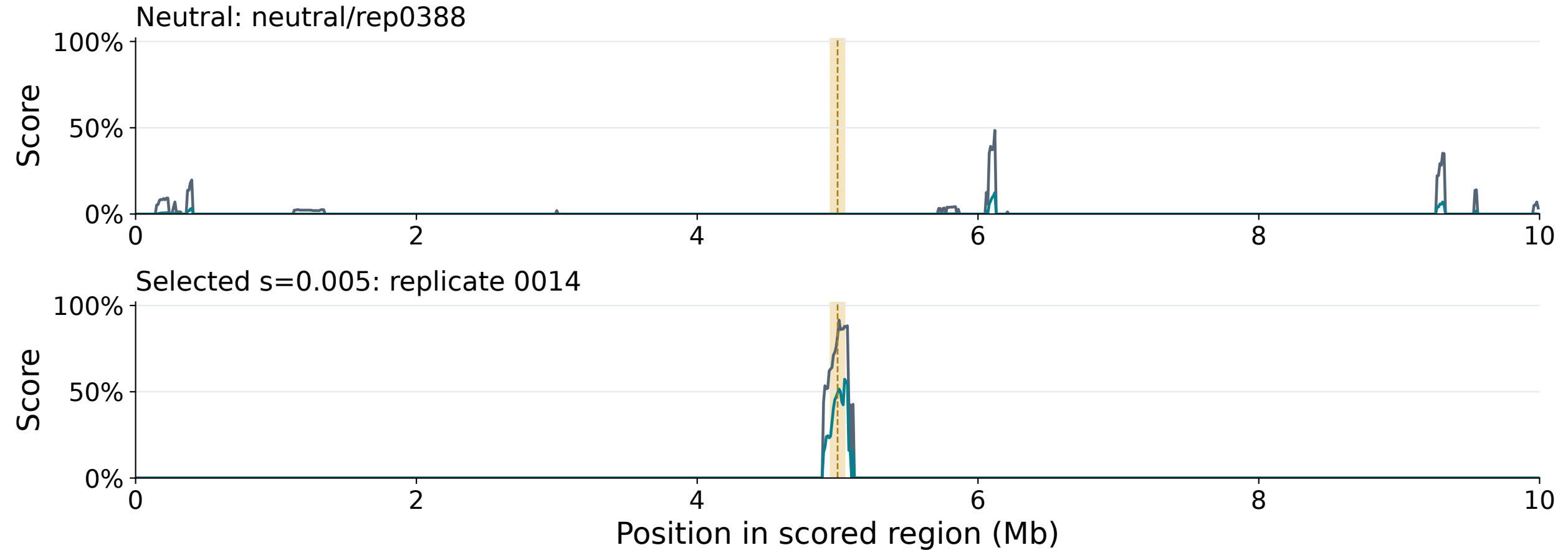
Nominal $p \leq 0.01$



A peak elsewhere does not invalidate the focal region

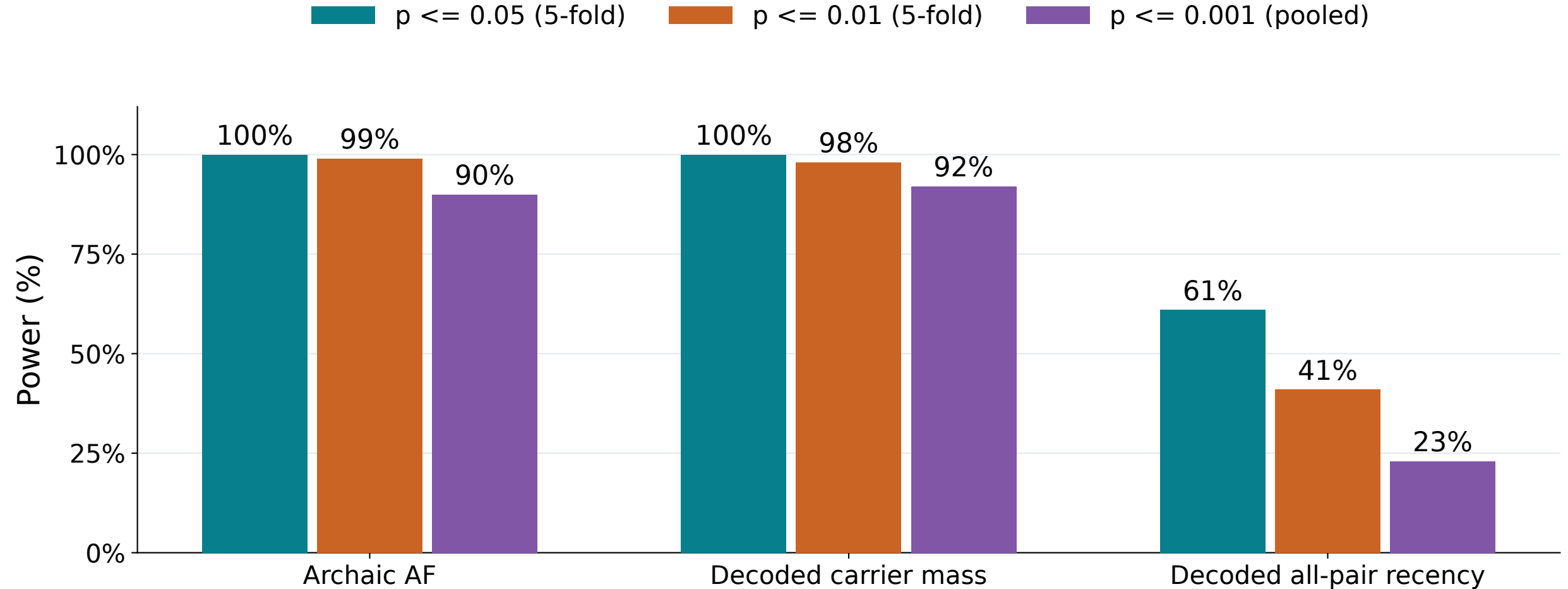
Illustrative examples selected by fixed rules; not an estimate of power

— Archaic AF — Decoded carrier mass



Stringency reduces decoded detection power

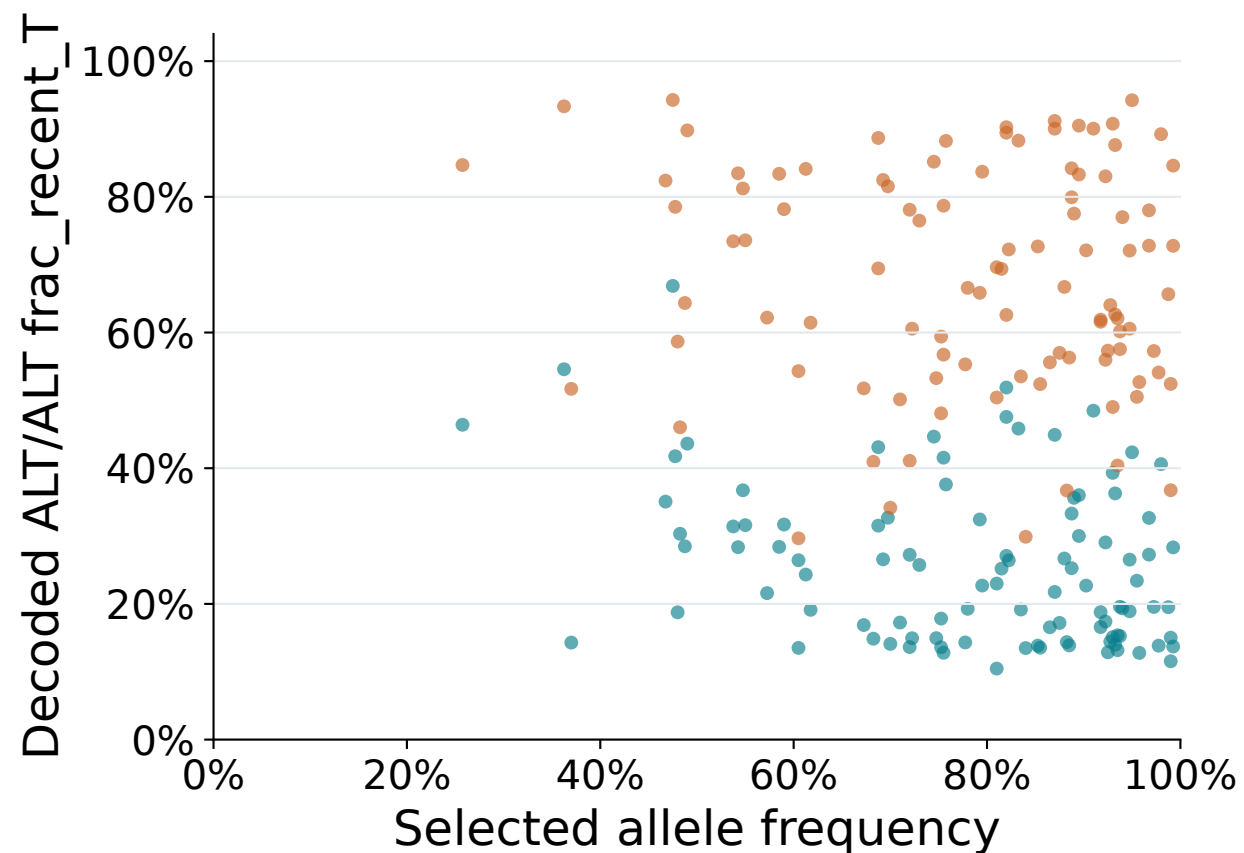
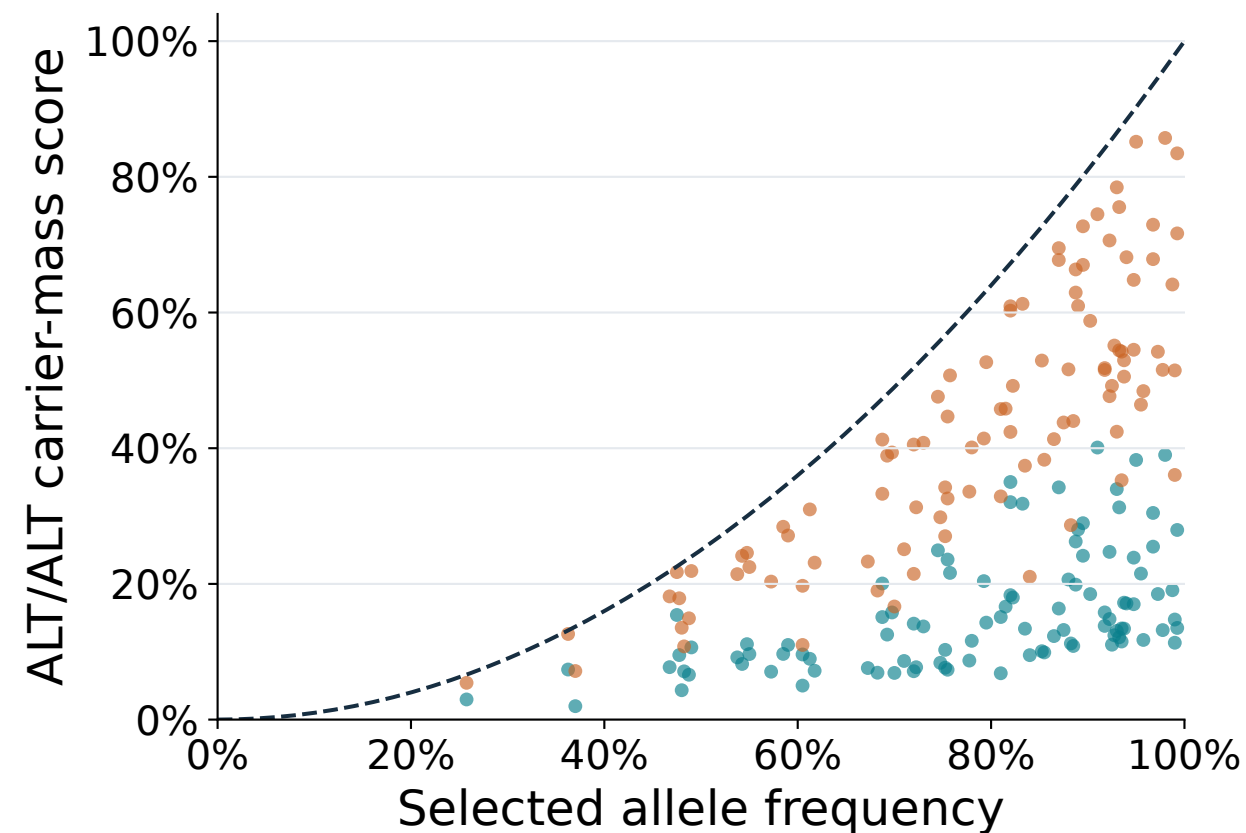
Gamma-SMC | $s = 0.005$; $T = 50$ kya | $p=0.001$ is a pooled-rank sensitivity analysis



Decoded carrier mass combines frequency and recency

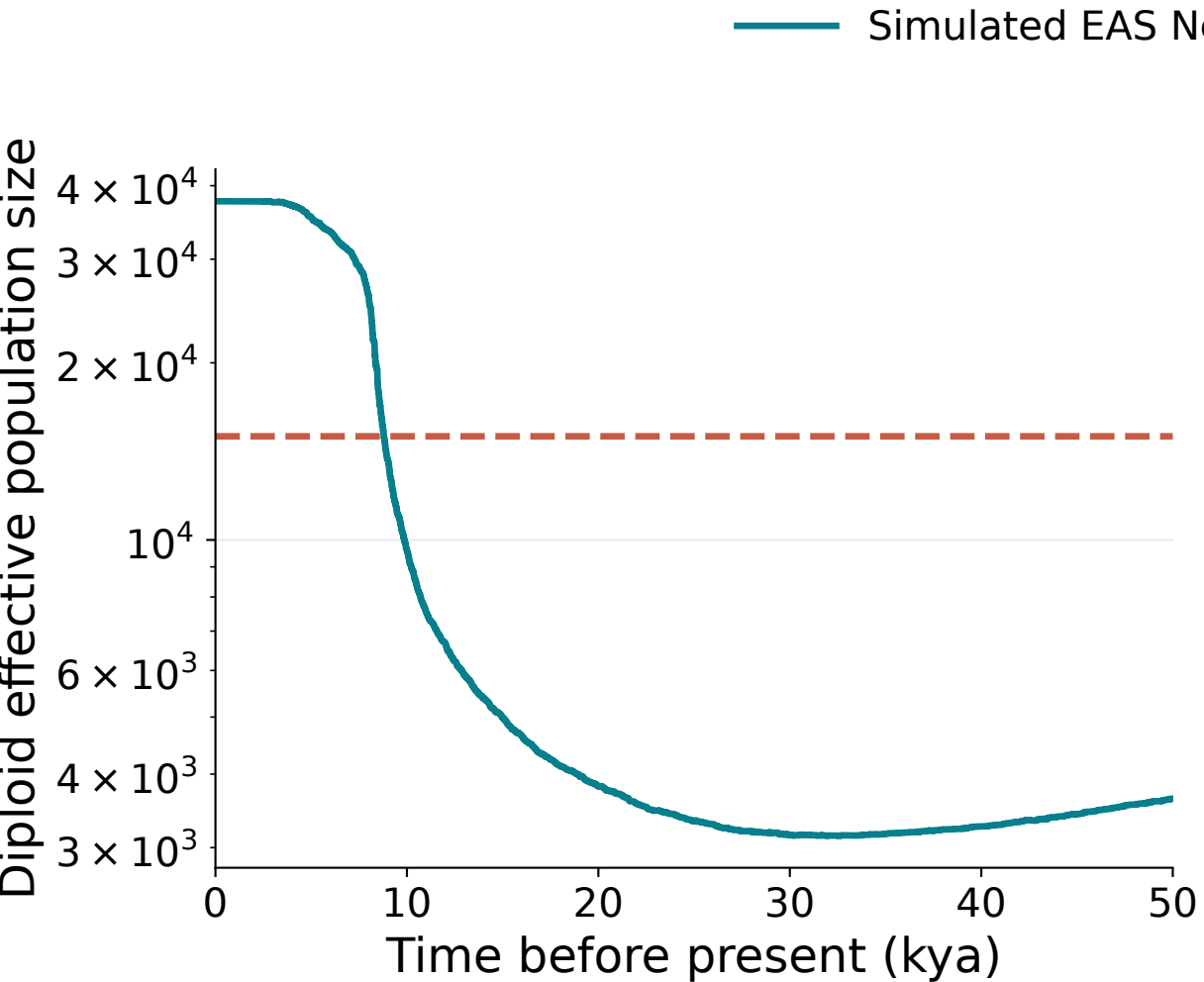
Gamma-SMC | 100 selected regions at $s = 0.005$ | each dot is one replicate

● T=20 kya ● T=50 kya ----- AF squared (approximation)



Demography is shared by neutral and selected simulations

The generative EAS history varies with time; the current decoder uses a fixed Ne



Archaic branch

Split: 700 kya

Baseline Ne: 3,600

Pulse bottleneck: Ne=10
for 100 generations

Mutation rate: 1.25×10^{-8}

Recombination rate: 1×10^{-8}

Generation time: 25 years

Gamma-SMC recency among archaic ALT/ALT pairs

Decoded focal TMRCA | pairs pooled within each class | $s=0.005$ selected versus neutral

